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SEAT No. _____



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SARDAR PATEL UNIVERSITY

B.C.A. (SEMESTER- 3)

US03CBCA23 || Object Oriented Programming -I

Date: 06-10-2025, Monday

Time: 2.00 pm – 5.00 pm

Marks: 70

Q-1 **Select the correct option from the following questions.** **(10)**

- 1) A _____ is a collection of objects of similar type.
a) Data b) Object c) Class d) Function
- 2) _____ is the developer of C++.
a) Dannis Ritchie b) Ivan Byross c) Bjarne Strastrup d) James Gosling
- 3) _____ means ability to take more than one form.
a) Inheritance b) Encapsulation c) Polymorphism d) Data hiding
- 4) For string manipulation _____ header file is included.
a) iostream.h b) ctype.h c) string.h d) conio.h
- 5) _____ is a selection control structure.
a) if b) switch c) if ... else d) All of Above
- 6) _____ is a user define data type.
a) Function b) Variable c) Class d) Operator
- 7) By default, main(.) in C++ returns _____.
a) void b) char c) int d) char*
- 8) Defining function with same prototype in base class & derived class is known
a) Function Overriding b) Function Overloading
c) Inheritance d) Inline function
- 9) The correct function name for overloading the addition (+) operator is
a) operator+ b) +operator c) operator : + d) operator(+)
- 10) Which of the following declarations are illegal?
a) Char str2 = "abc" b) Void *ptr
c) Char *str1 = "xyz" d) Const *int p1

Q-2 **Short Questions. (Attempt any ten)** **(20)**

- 1) Define encapsulation as concept of OOP.
- 2) List out any four header file in C++.
- 3) Differentiate variable and constant.
- 4) Define nameless and arrays of objects.
- 5) List the characteristics of member function.
- 6) Define destructor and give its example.

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(P.T.O.)

- 7) Define inheritance & state two advantages of it.
- 8) What is the difference between private & protected members of the class?
- 9) What is inline function? Give an example of it.
- 10) What are input and output stream?
- 11) What is an operator function? Describe the syntax of an operator function.
- 12) List the operators that can't be overloaded?

- Q-3** **A** Explain the concept of class & object with example. (05)
 B Explain different data types available in C++. (05)

OR

- Q-3** **A** Explain different operator available in C++. (05)
 B Explain features (characteristics) of OOP in detail. (05)

- Q-4** **A** What do you mean by constructors? What is the use of constructors in C++? (05)
 B Explain basic I/O in C++ with proper example. (05)

OR

- Q-4** **A** Explain Copy constructor with example. (05)
 B Explain all operations on Arrays. (05)

- Q-5** **A** Explain multiple & multilevel inheritance with suitable example. (05)
 B Explain the concept of friend function with suitable example. (05)

OR

- Q-5** **A** Discuss briefly – virtual function (05)
 B Discuss inheritance in private mode & its effects on accessibility of base-class members. (05)

- Q-6** Explain unary operator overloading using member function & friend function with proper example. (10)

OR

- Q-6** What is file mode? Describe the various file mode operations available. Describe the various classes available for file operations. (10)

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